

Aaron Sher

Software Engineer & Data Scientist · Python · Backend Systems · Data Science · MLOps · ML Engineering · EU Citizen
[Portfolio](#) · [LinkedIn](#) · asher897@gmail.com

PROFILE

Software Engineer and Data Scientist with 4+ years in high-performance backend systems and a year in data engineering and ML. Strong record building scalable microservice infrastructure for a major bank and end-to-end data pipelines for healthcare analytics. B.Sc Honours research in medical image segmentation using deep learning.

EXPERIENCE

Data Scientist | **Insight (Healthcare Analytics Consultancy)** *Aug 2025 – Present*

- Solely designed and built a modular ETL framework on Airflow and Polars — config-driven architecture where pipelines are assembled from reusable components rather than written end-to-end each time, replacing SQL transforms with in-memory processing.
- Built and deployed automated Airflow workflows for auditable, reproducible data preparation across healthcare client datasets.
- Led recovery pathway modelling for knee and hip arthroplasty patients using gradient-boosted trees and clustering to identify drivers of different clinical outcomes.
- Research accepted at the ISQUA 42nd International Conference.
- Conducted FWA (Fraud, Waste & Abuse) pattern detection across client datasets to support actuarial analysis.

Software Engineer | **BBD (contracted to Discovery Bank)** *Jan 2022 – Jul 2025*

- Owned the core backend framework underpinning the bank's microservice infrastructure (Python & C++); led framework migrations, critical fixes, and architectural modernisation across both legacy and modern services.
- Redesigned RabbitMQ integration end-to-end — ~50% latency reduction and halved CPU usage across multiple process flows.
- Sole engineer responsible for the legacy C++ codebase, developing new features and maintaining integration with modern Python services throughout.
- Designed an async Python worker process framework from scratch, eliminating synchronous bottlenecks and significantly improving throughput.
- Engineered cross-cluster intermediary broker services enabling zero-downtime migration between Kubernetes clusters with separate message brokers.
- Built a WhatsApp orchestration engine as part of a small team — managed all inbound and outbound message flows including media, AI chatbot integration, end-to-end encryption, API authorisation, bulk notification APIs, and agent dashboards for campaign management.
- Integrated WebSocket streaming between call-centre agents and AI services for real-time response delivery.
- Took on SRE responsibilities including CI/CD pipeline management, Kubernetes deployment oversight, and the full observability stack across Grafana, Prometheus, and OpenTelemetry.
- Mentored junior developers through code reviews, pair programming, and work assignment; helped establish patterns adopted by the wider team.

Managing Director | **Habonim Dror (National Youth Movement)** *2021*

- Led the national executive across Johannesburg and Cape Town, overseeing programmes, budgets, and a team of regional coordinators.

EDUCATION

B.Sc (Hons) Computer Science *2023 – 2024*

Coursework: Deep Learning, CV, NLP, RL, Data Analysis, Mathematical Foundations for ML, Robotics

Research: Bagging & Cross-Attention for Feature Context Extraction in Medical Image Analysis — improving domain transfer in segmentation tasks (COVID-19, TB).

B.Sc Computer Science & Economics *2018 – 2020*

SKILLS & TECHNOLOGIES

ML / AI PyTorch · TensorFlow · scikit-learn · Hugging Face · LLMs · RAG · LangChain · LlamaIndex · Weights & Biases · MLflow · NLP · Computer Vision · Reinforcement Learning

CV & Imaging OpenCV · Albumentations · torchvision · PIL/Pillow · scikit-image · Medical image analysis

Data Polars · pandas · NumPy · SciPy · Airflow · SQL · Jupyter · Data pipelines · Feature engineering · Data analysis

Backend Python · C++ · C# · Java (Spring) · FastAPI · Flask · Tornado · REST API design · Backend architecture · Microservices · Async frameworks · Celery · Redis · RabbitMQ · SQLAlchemy · pyodbc · pytest

Infra / Cloud Docker · Kubernetes · Azure (App Services, AKS, Functions, ML Studio) · CI/CD · Git · Linux · Bash · Grafana · Prometheus · OpenTelemetry · Helm · SRE